South Africa, step 2 (2026-11-02)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals
1	20263016	5000	0.275	−40982.9	−40967.8	0
2	20263017	5000	0.276	−40984.6	−40970.8	0
3	20263018	5000	0.275	−40980.6	−40968.5	0
4	20263019	5000	0.274	−40984.5	−40970.9	0

file
mcmc_south_africa_step2_chain1_2026-10-31.Rdata
mcmc_south_africa_step2_chain2_2026-10-31.Rdata
mcmc_south_africa_step2_chain3_2026-10-31.Rdata
mcmc_south_africa_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

age.30.to.39.transmission.multiplier.3	age.20.to.29.transmission.multiplier.0
1.958	1.856
trate.1	age.20.to.29.transmission.multiplier.1
1.712	1.568
male.engagement.multiplier	age.50.and.over.transmission.multiplier.3
1.507	1.478
log.OR.engagement.slope	female.to.male.multiplier
1.443	1.434
trate.3	log.OR.testing.intercept
1.390	1.372
age.15.to.19.transmission.multiplier.3	trate.4
1.335	1.331
log.OR.suppression.slope	age.40.to.49.transmission.multiplier.1
1.278	1.275
age.30.to.39.transmission.multiplier.0	age.50.and.over.transmission.multiplier.1
1.267	1.247
age.30.to.39.transmission.multiplier.1	age.15.to.19.transmission.multiplier.1
1.247	1.235
age.20.to.29.transmission.multiplier.3	hiv.specific.mortality.rates.1
1.225	1.215
age.25.to.49.hiv.mortality.multiplier.1	age.30.to.39.transmission.multiplier.2
1.151	1.130
age.20.to.29.transmission.multiplier.2	age.50.and.over.transmission.multiplier.2
1.120	1.104
age.45.to.64.mortality.slope.multiplier.male	
1.103	

Merged simset:

location	simulations	total.weight	file
South Africa	1000	1000	simset_south_africa_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 51 of 72 targets flagged (worst 30 shown; all in <location>_fit_check.csv)

data.type	age	sex	years.with.data	pct.years.in.sim.95	median.sim.vs.data	last.year	last.sim.vs.data
hiv.mortality	0-14	both	26	15	x0.43	2025	x0.10
incidence	50 and over	male	26	12	x0.46	2025	x1.29
hiv.mortality	25-49	male	26	0	x0.51	2025	x0.04
hiv.mortality	25-49	both	26	0	x0.51	2025	x0.05
hiv.mortality	25-49	female	26	0	x0.52	2025	x0.05
hiv.mortality	All ages	male	26	8	x0.52	2025	x0.07
hiv.mortality	All ages	both	26	8	x0.54	2025	x0.09
hiv.mortality	15-24	female	26	27	x0.52	2025	x0.14
hiv.mortality	All ages	female	26	12	x0.55	2025	x0.10

South Africa step 2: likelihood

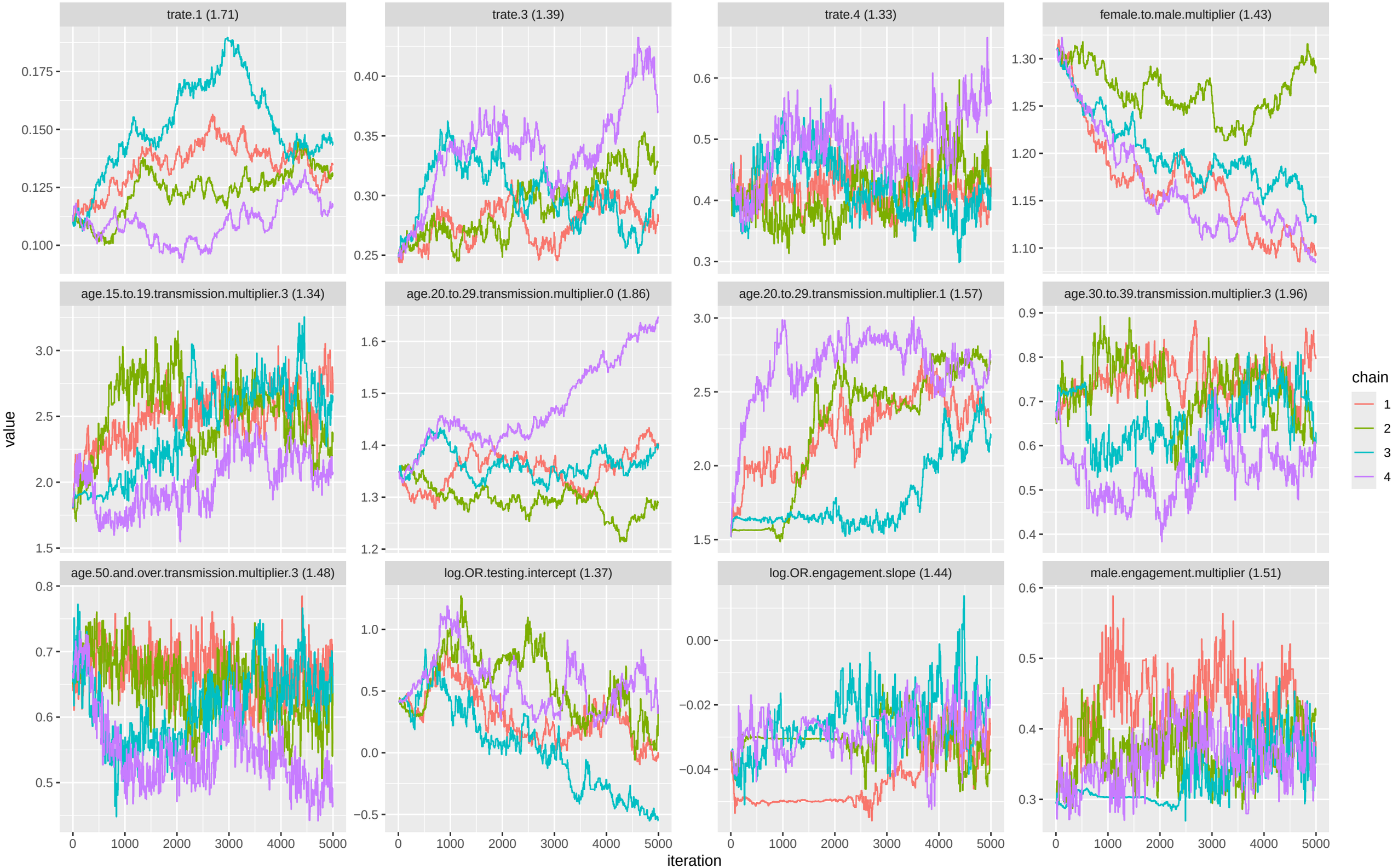
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

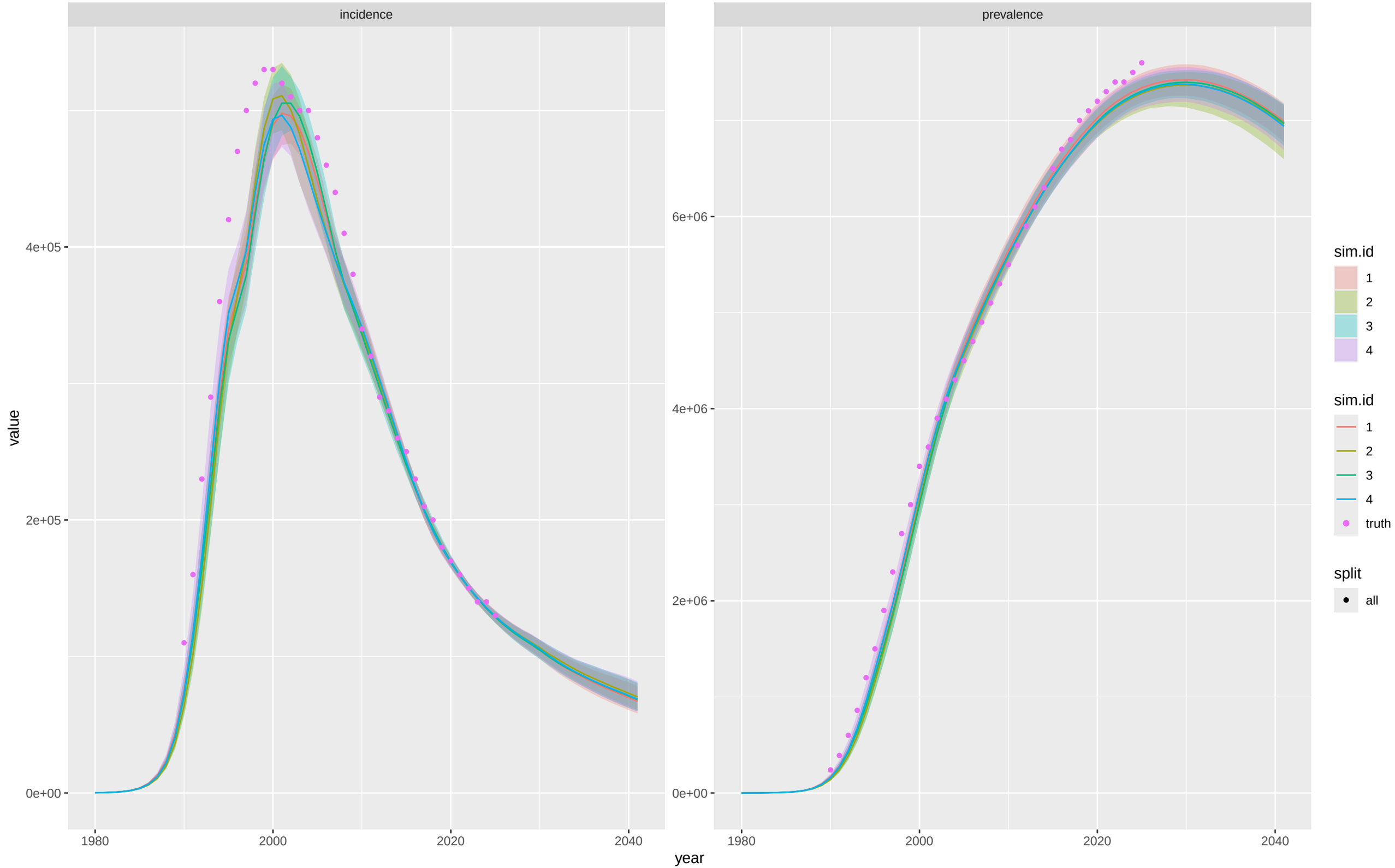
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

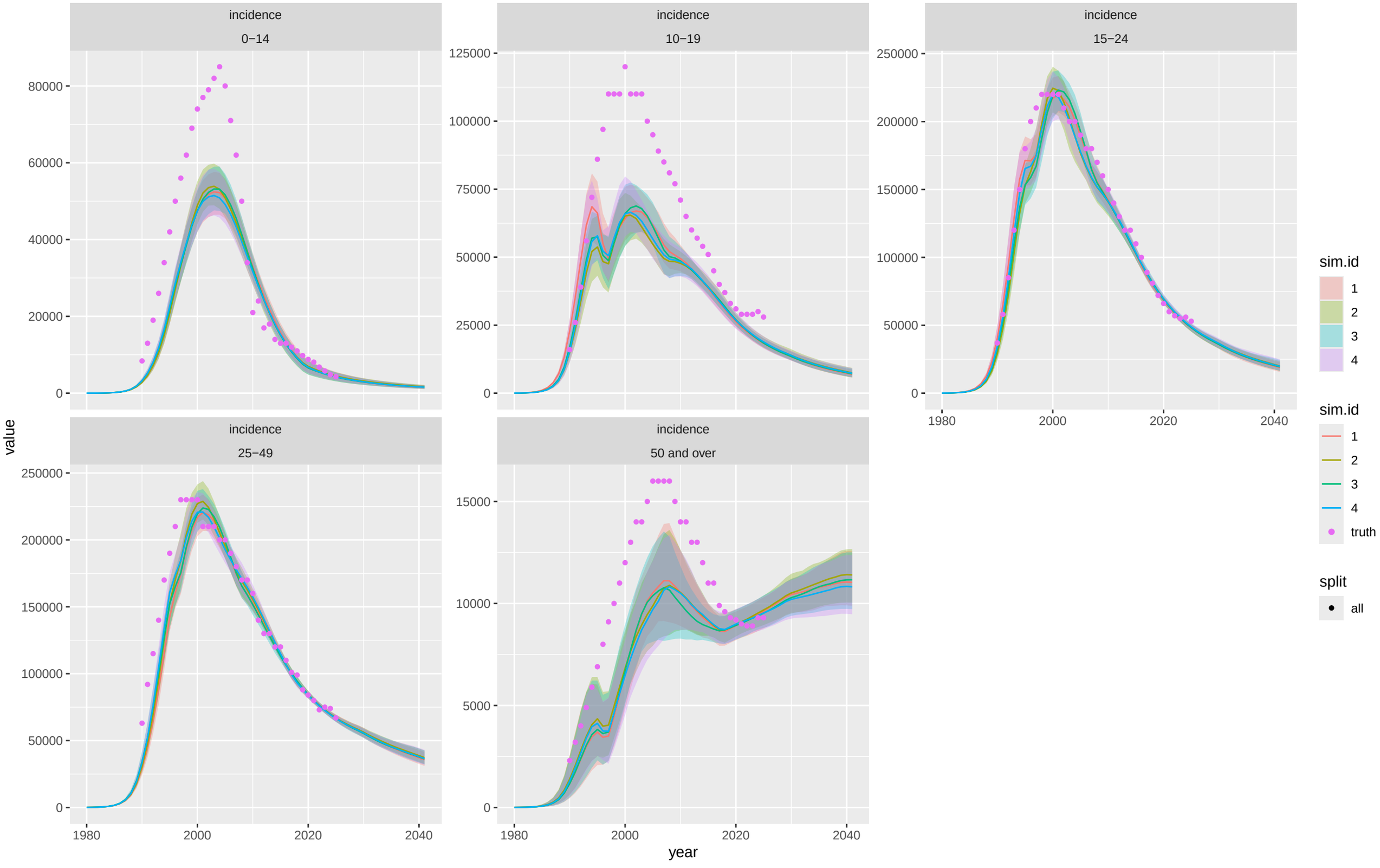
South Africa step 2: trace



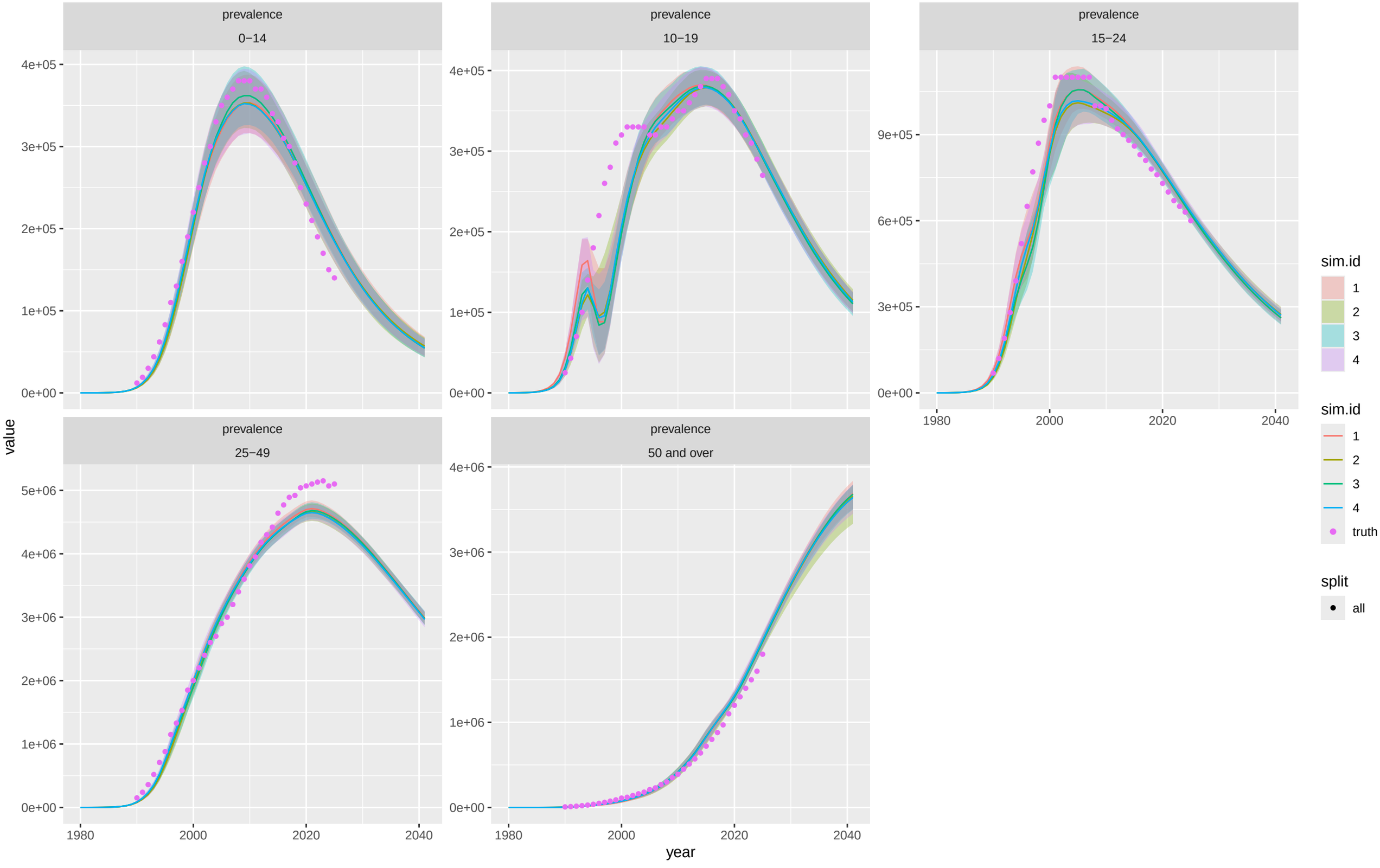
South Africa step 2: fit (one color per chain)



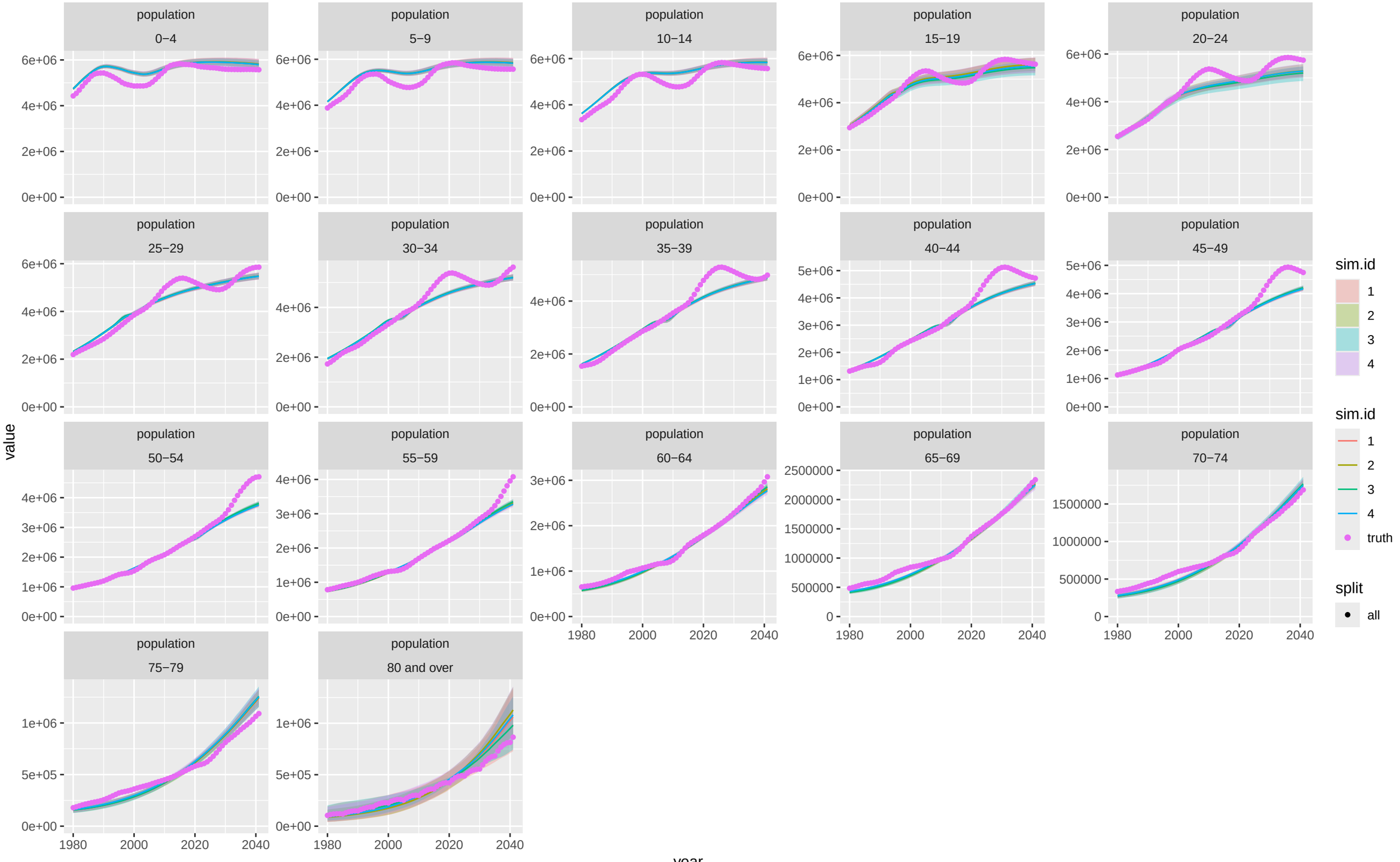
South Africa step 2: incidence.by.age (one color per chain)



South Africa step 2: prevalence.by.age (one color per chain)



South Africa step 2: population.by.age (one color per chain)



South Africa step 2: cascade (one color per chain)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

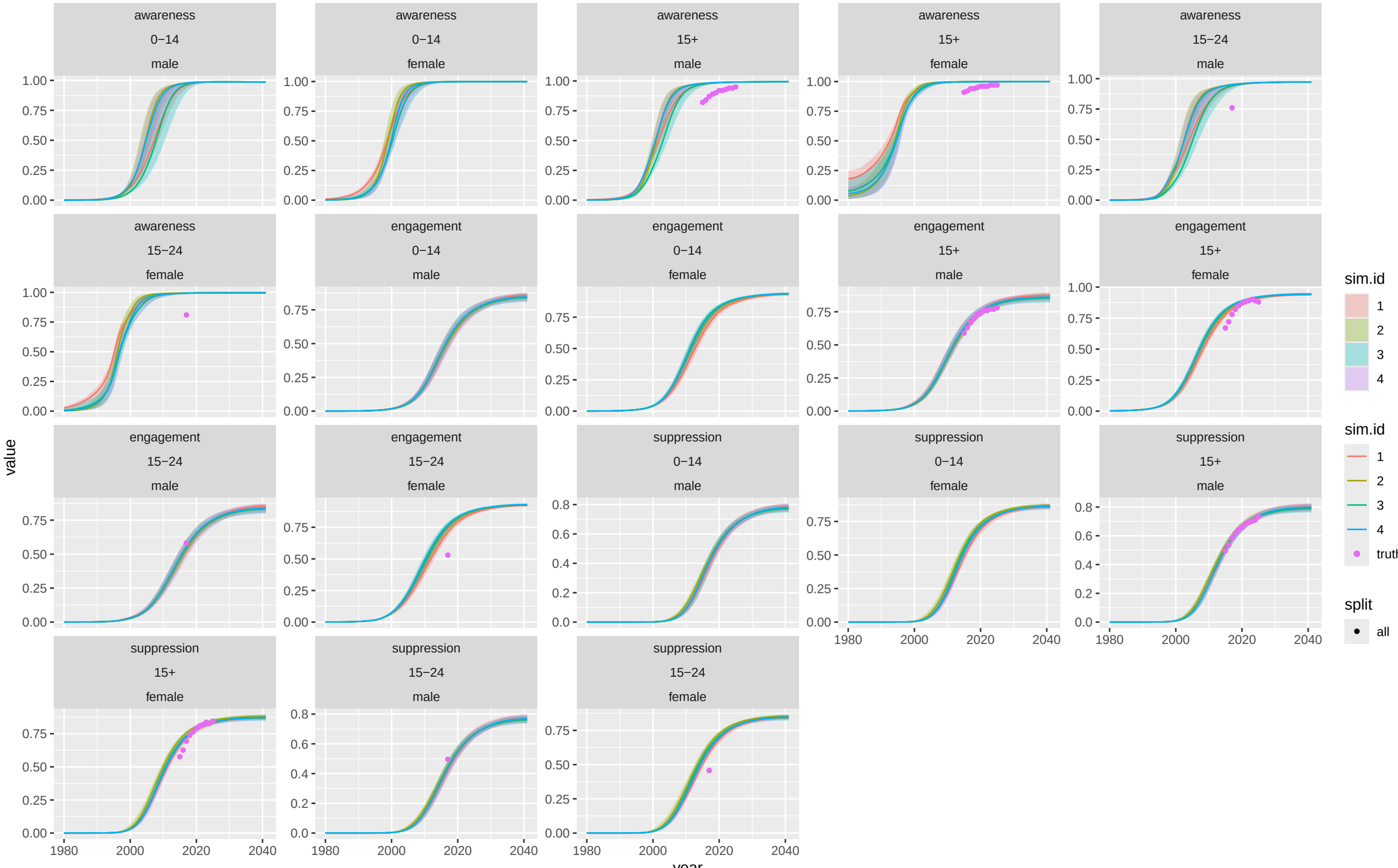
1980
2000
2020
2040

sim.id
1
2
3
4

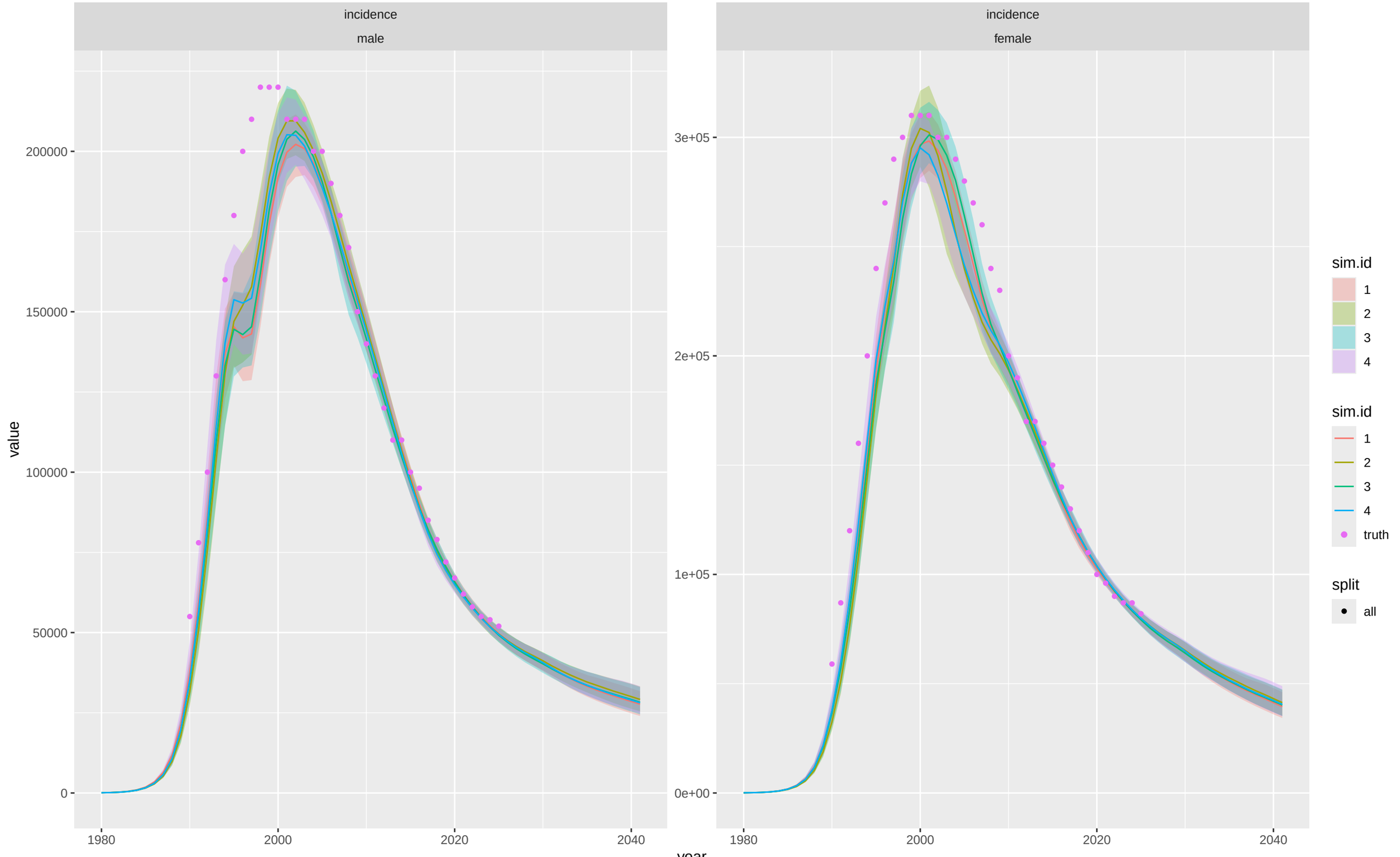
sim.id
1
2
3
4
truth

split
all

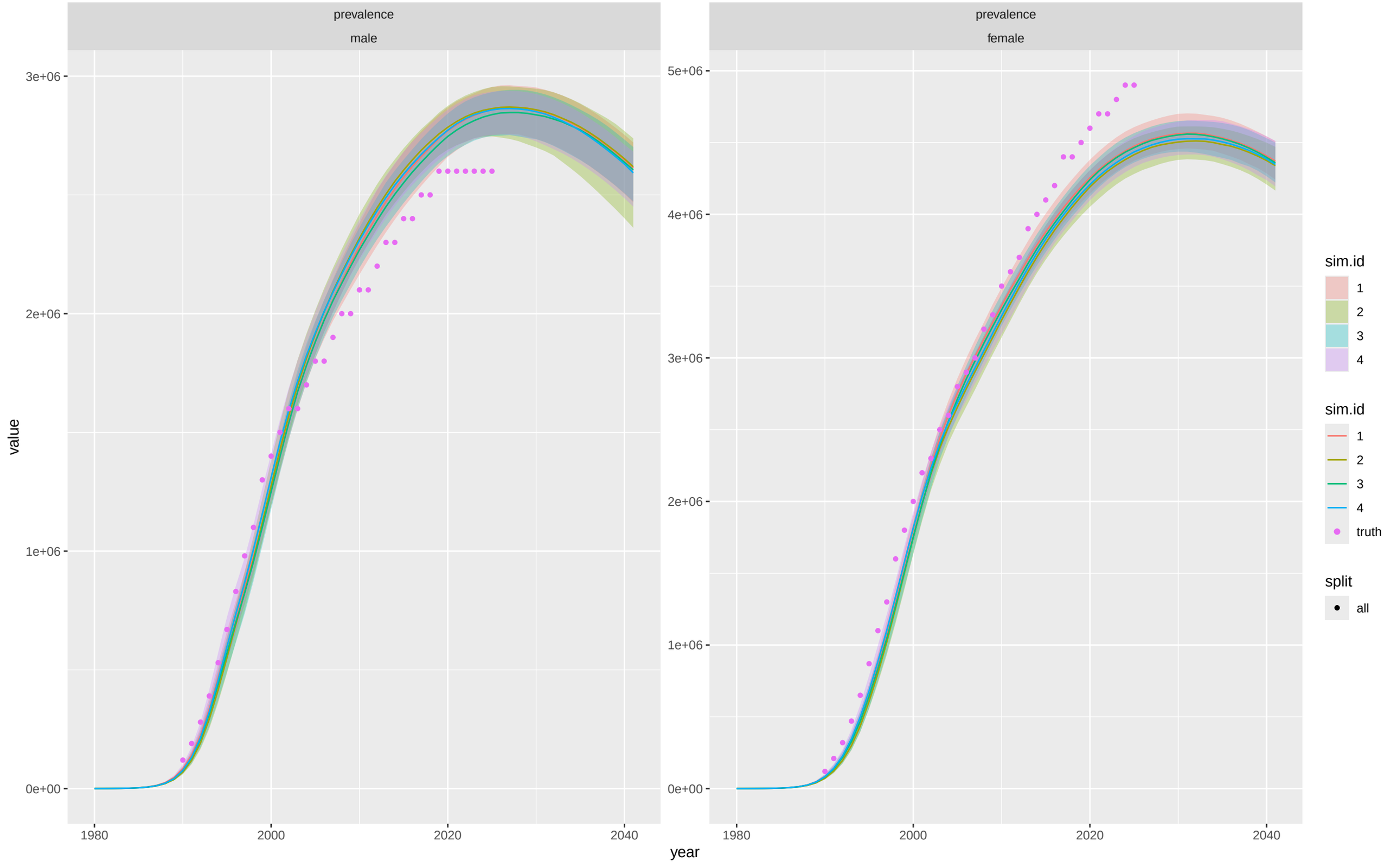
South Africa step 2: cascade.by.age.sex (one color per chain)



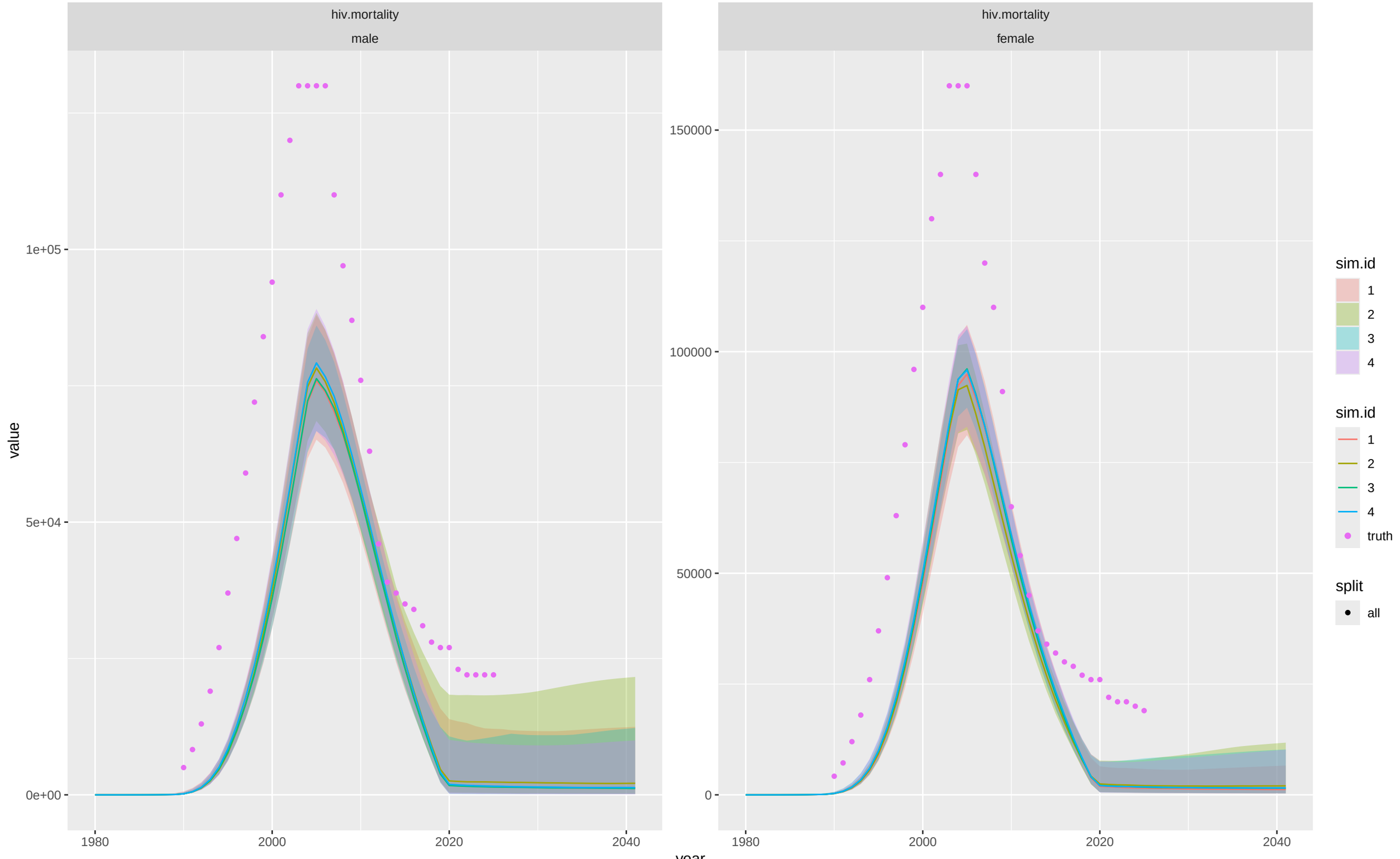
South Africa step 2: incidence.by.sex (one color per chain)



South Africa step 2: prevalence.by.sex (one color per chain)



South Africa step 2: hiv.mortality.by.sex (one color per chain)



South Africa step 2: incidence.by.age.sex (one color per chain)

incidence 0−14 male

incidence 0−14 female

incidence 10−19 male

incidence 10−19 female

incidence 15−24 male

incidence 15−24 female

incidence 25−49 male

incidence 25−49 female

incidence 50 and over male

incidence 50 and over female

value

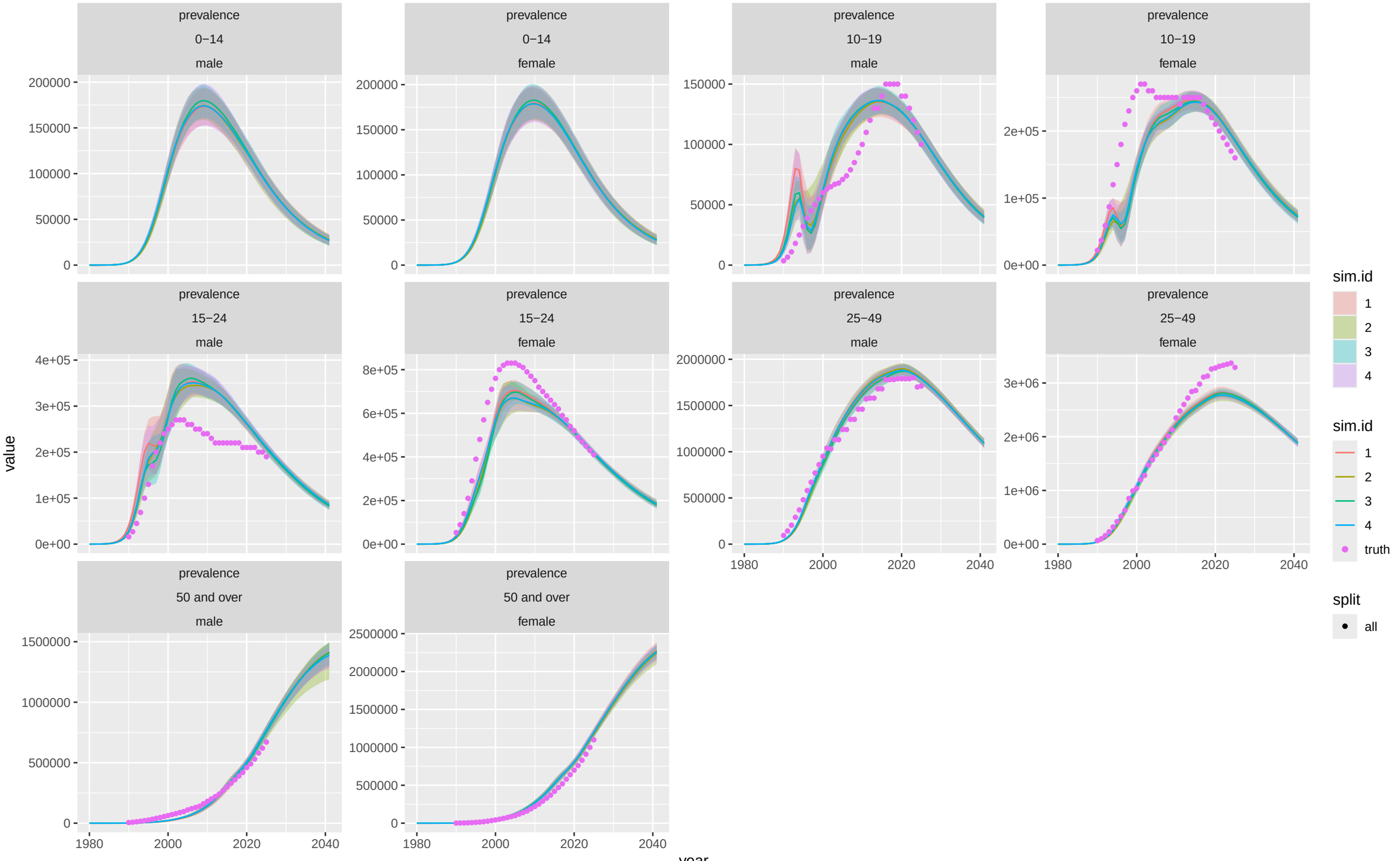
year

sim.id
1
2
3
4

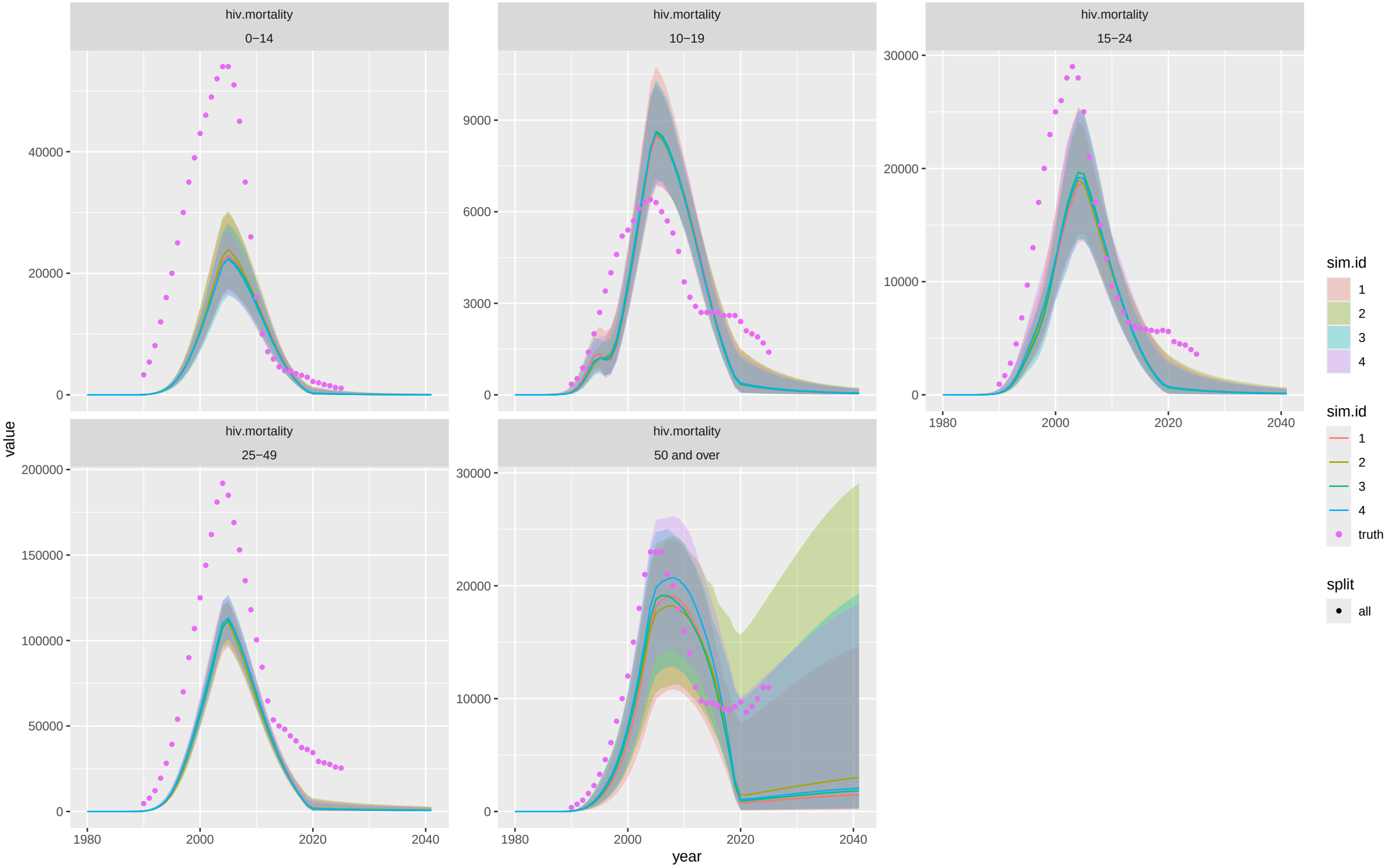
sim.id
1
2
3
4
truth

split
all

South Africa step 2: prevalence.by.age.sex (one color per chain)



South Africa step 2: hiv.mortality.by.age (one color per chain)



South Africa step 2: hiv.mortality.by.age.sex (one color per chain)

hiv.mortality 0–14 male

hiv.mortality 0–14 female

hiv.mortality 10–19 male

hiv.mortality 10–19 female

hiv.mortality 15–24 male

hiv.mortality 15–24 female

hiv.mortality 25–49 male

hiv.mortality 25–49 female

hiv.mortality 50 and over male

hiv.mortality 50 and over female

value

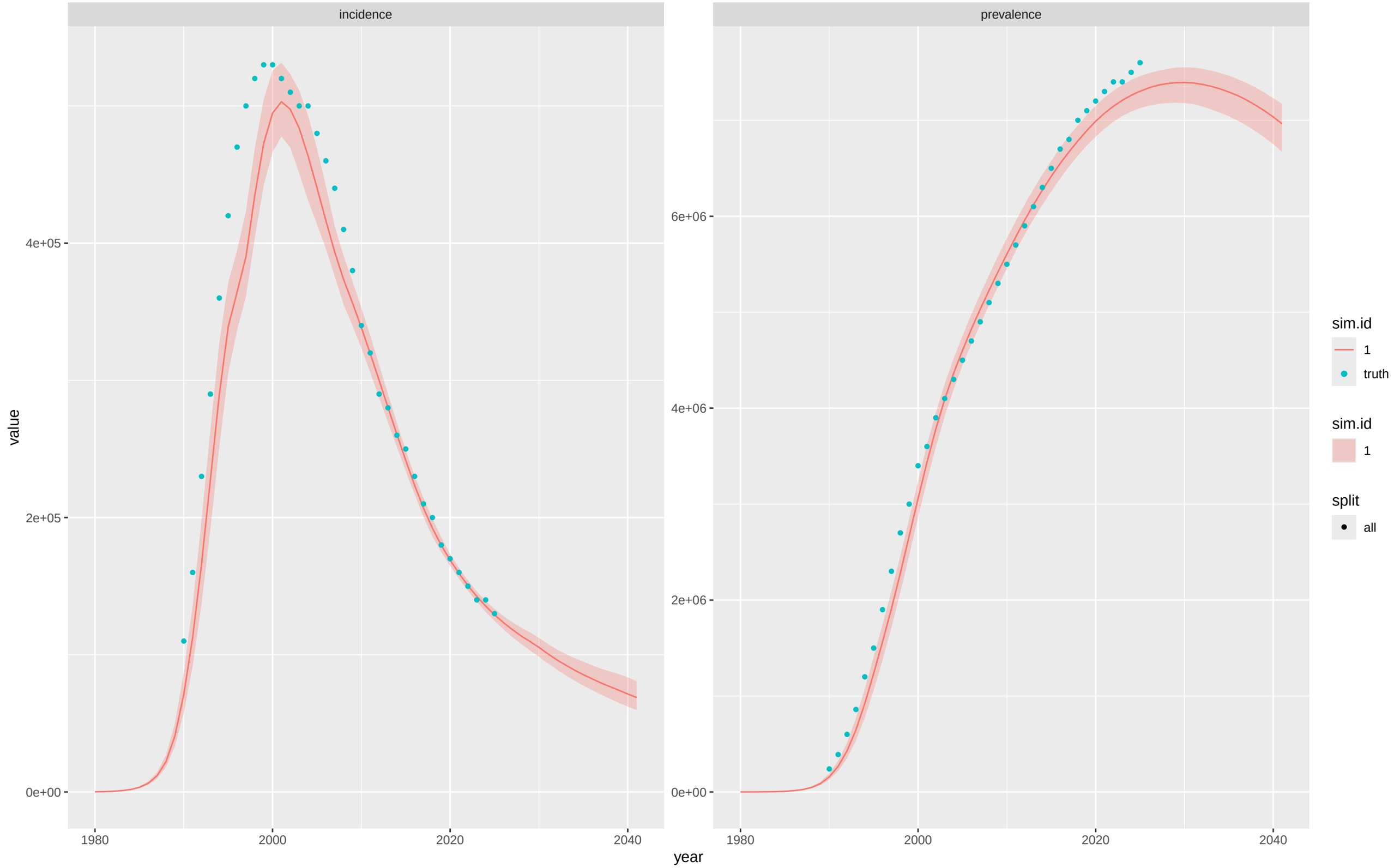
year

sim.id
1
2
3
4

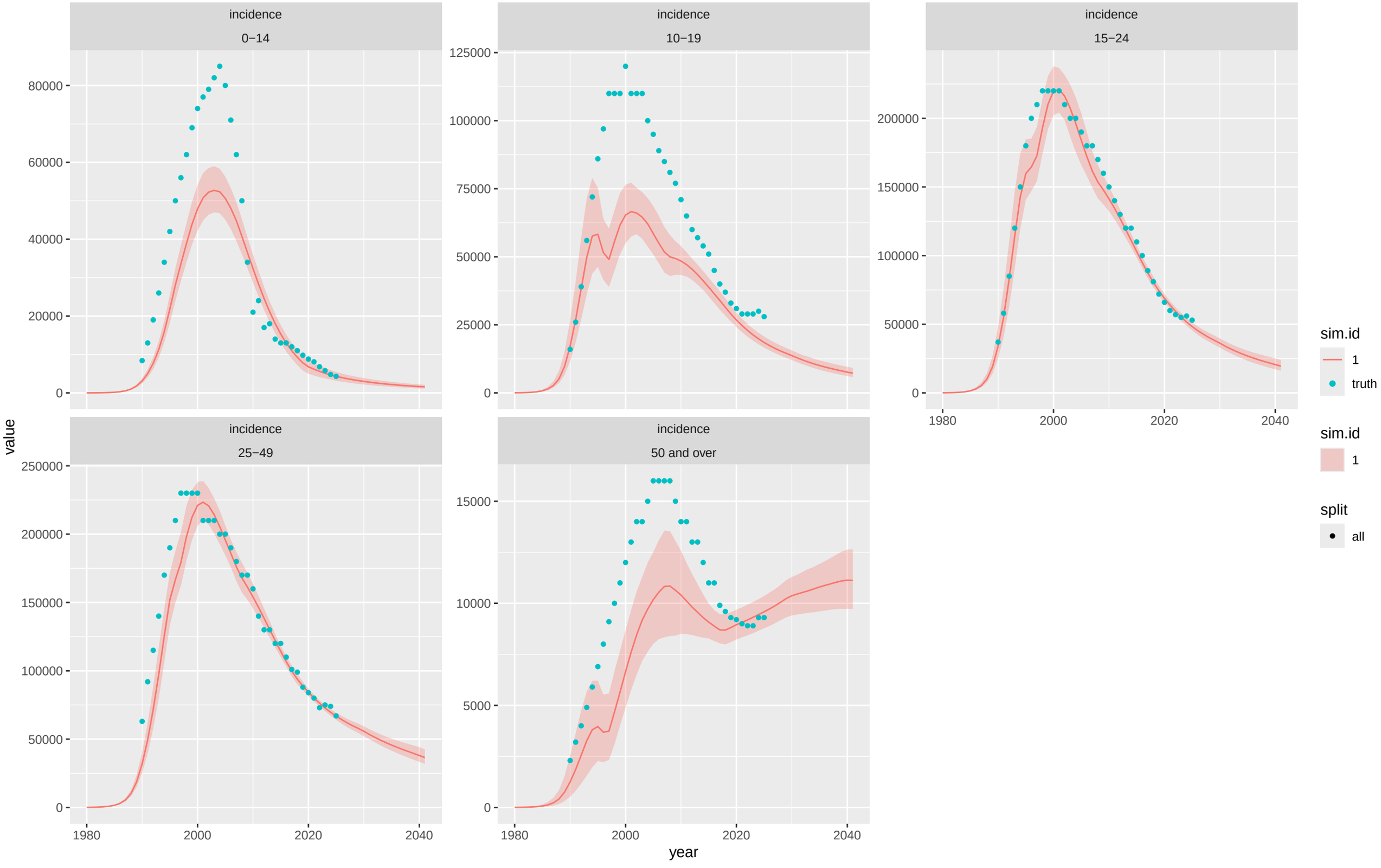
sim.id
1
2
3
4
truth

split
all

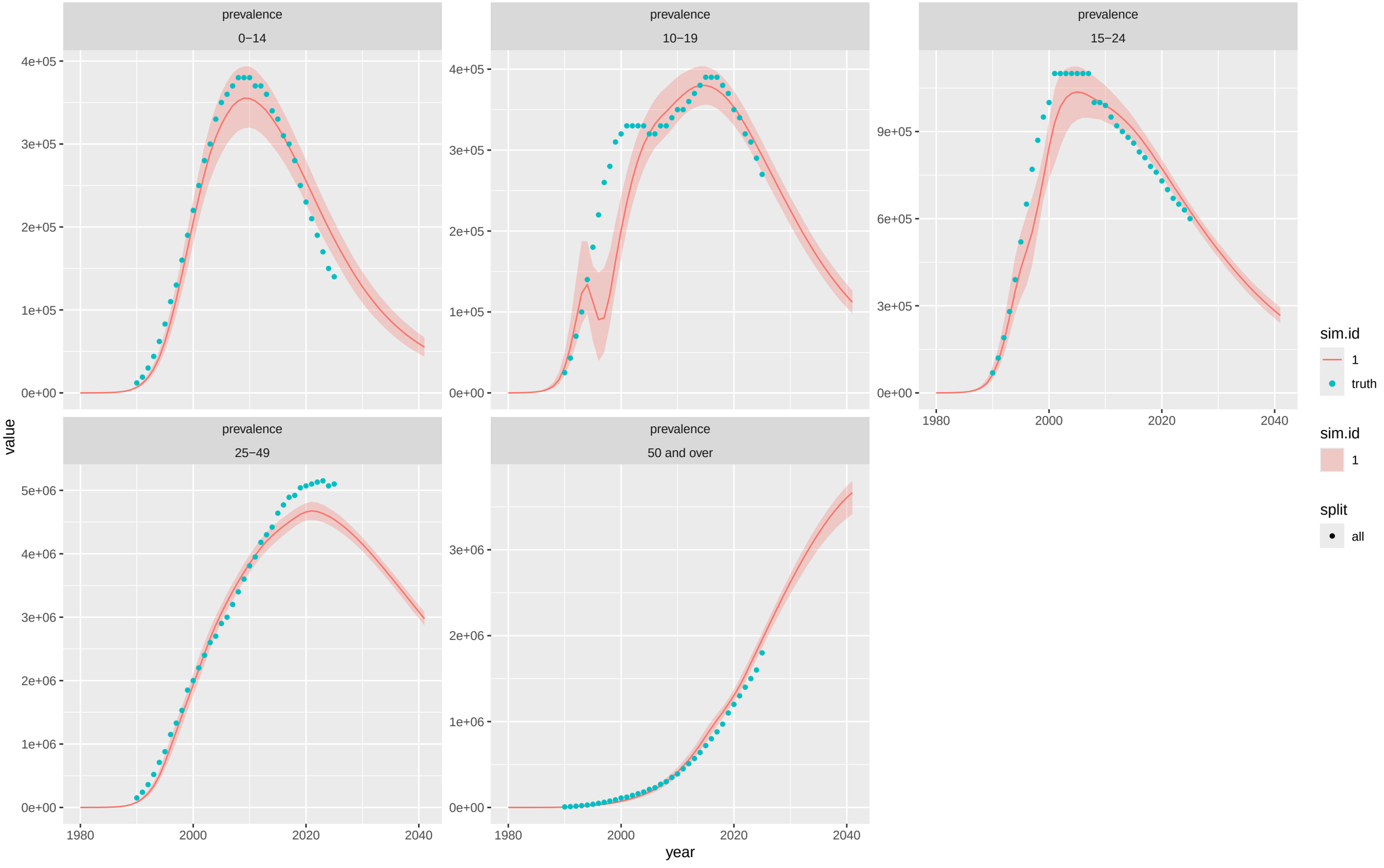
South Africa: fit (merged Step 2 simset)



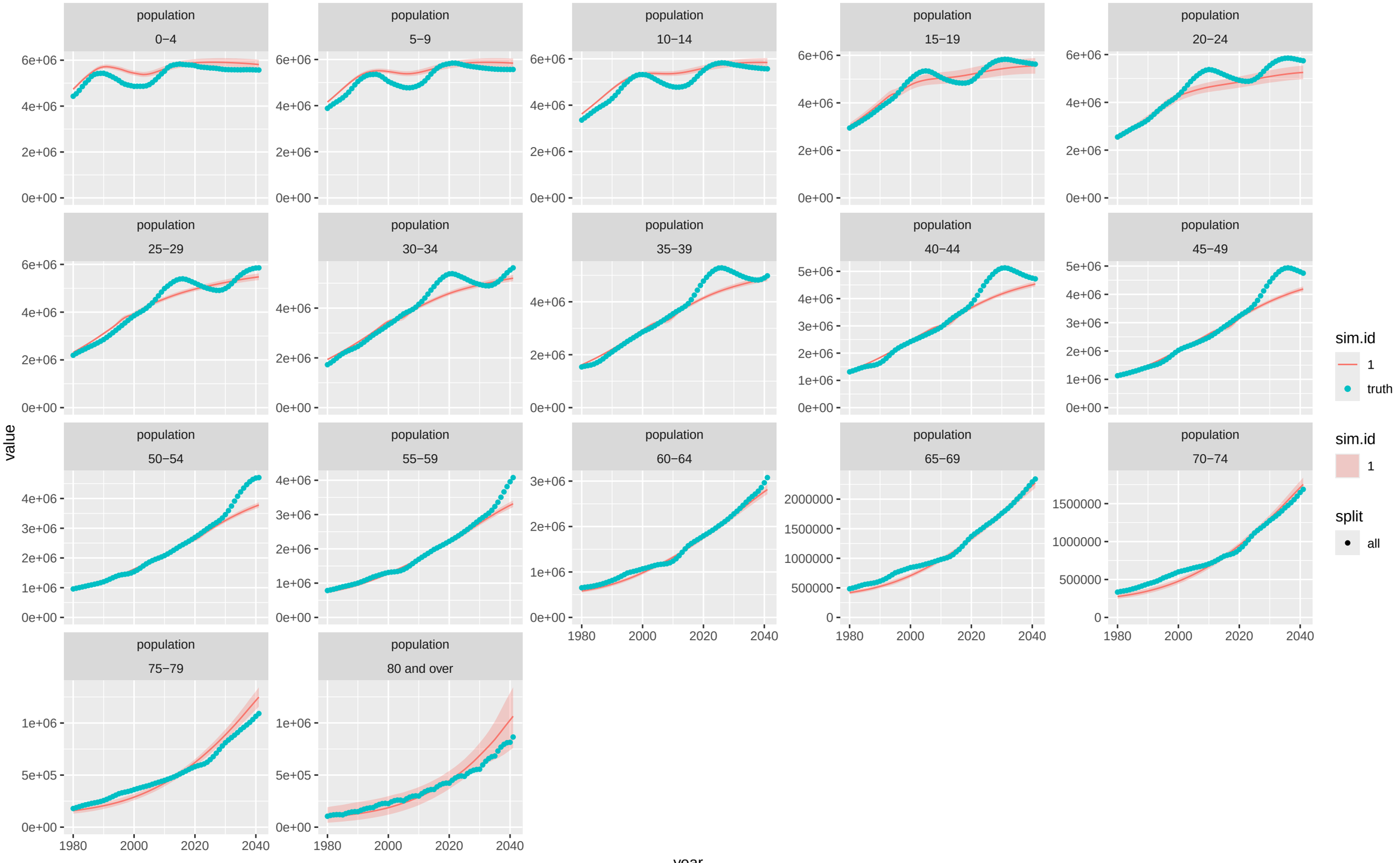
South Africa: incidence.by.age (merged Step 2 simset)



South Africa: prevalence.by.age (merged Step 2 simset)



South Africa: population.by.age (merged Step 2 simset)



South Africa: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

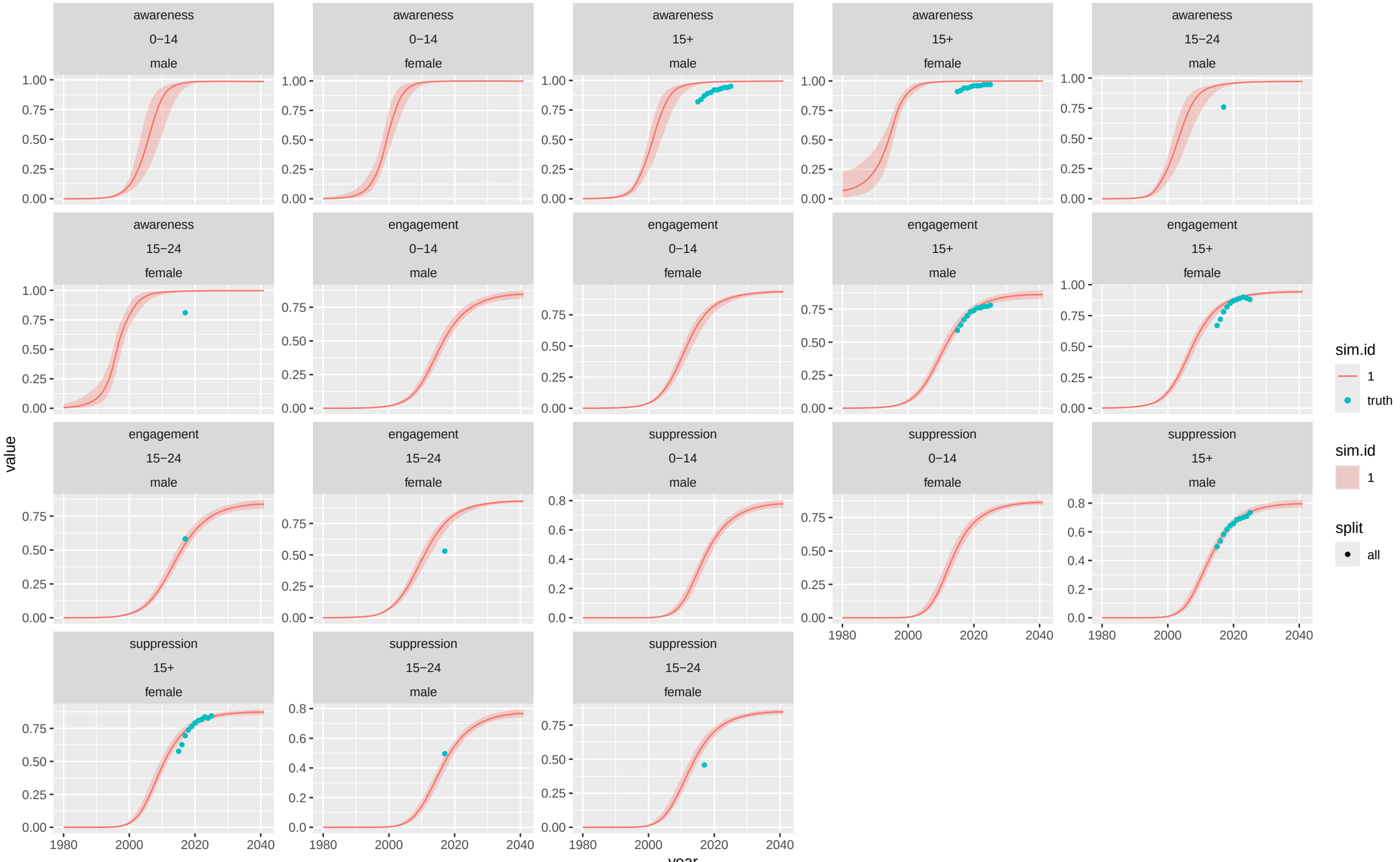
sim.id

1

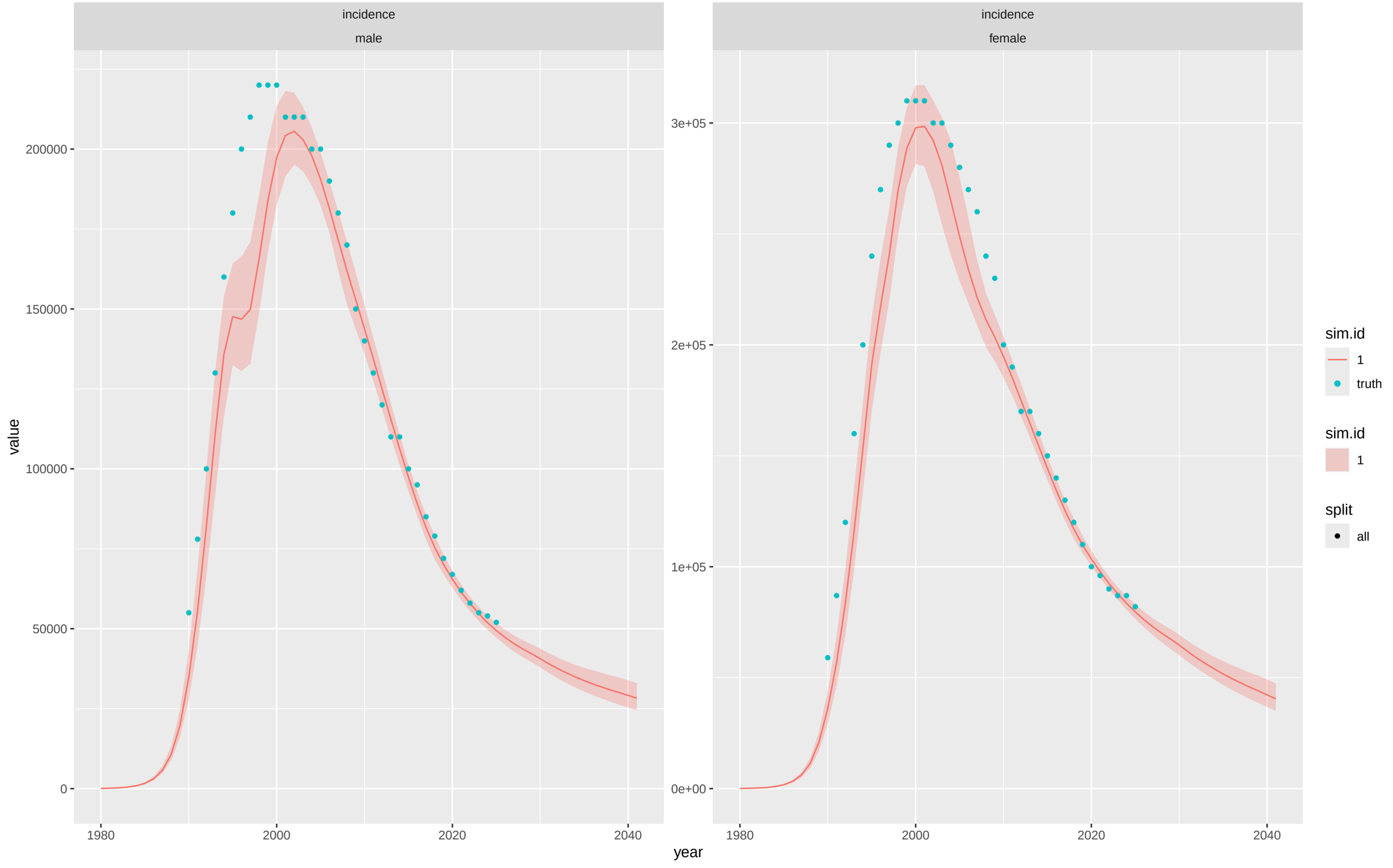
split

all

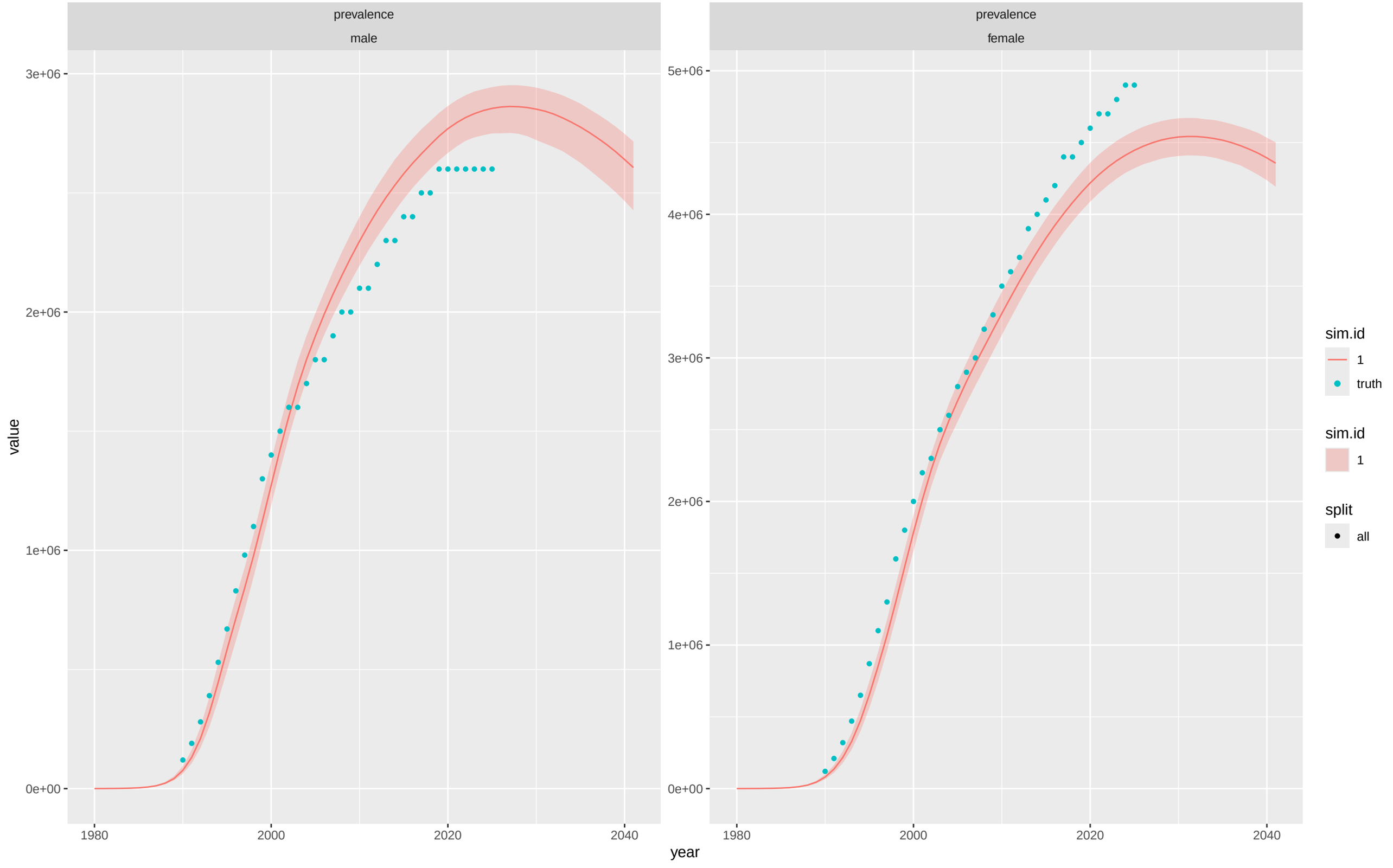
South Africa: cascade.by.age.sex (merged Step 2 simset)



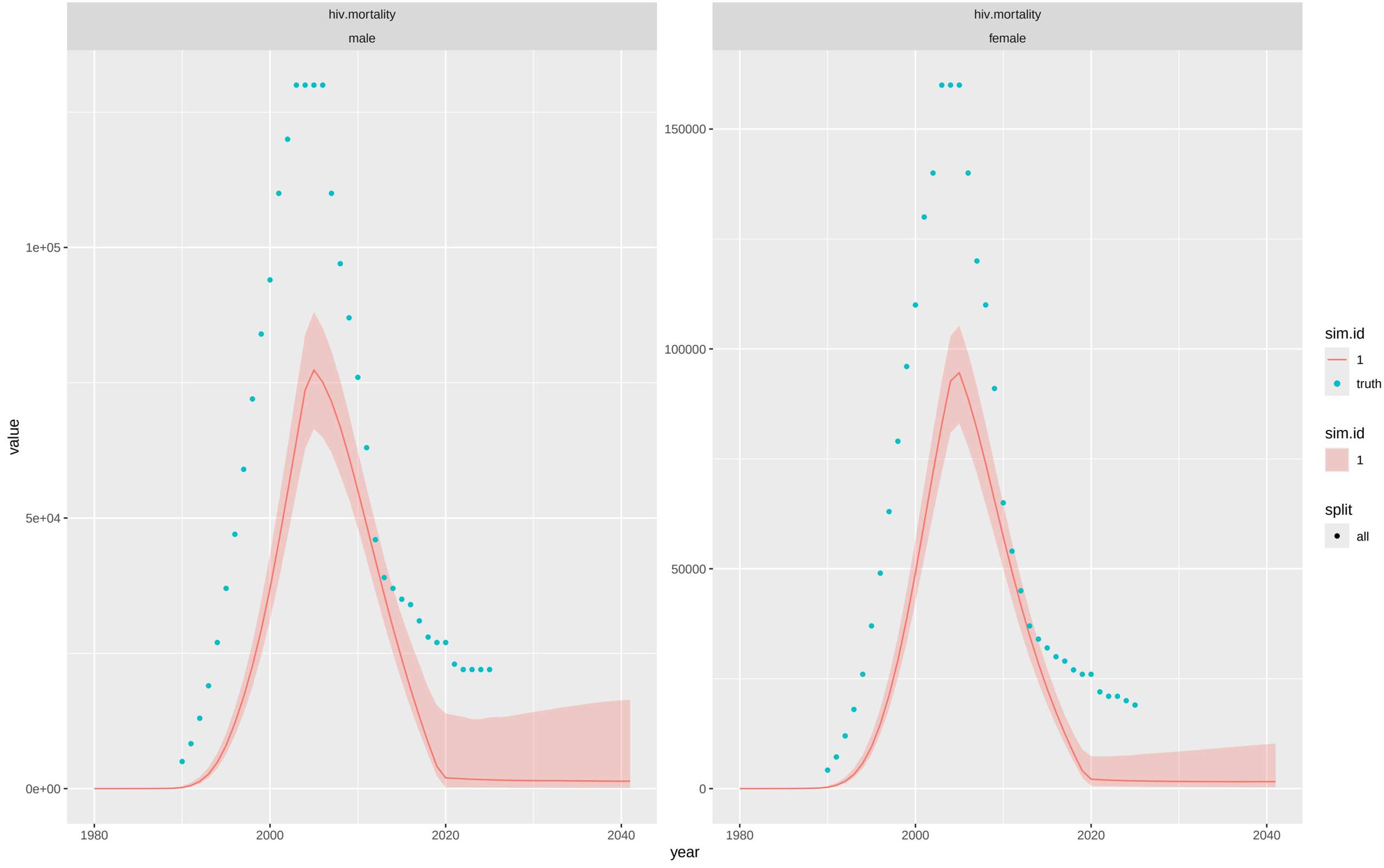
South Africa: incidence.by.sex (merged Step 2 simset)



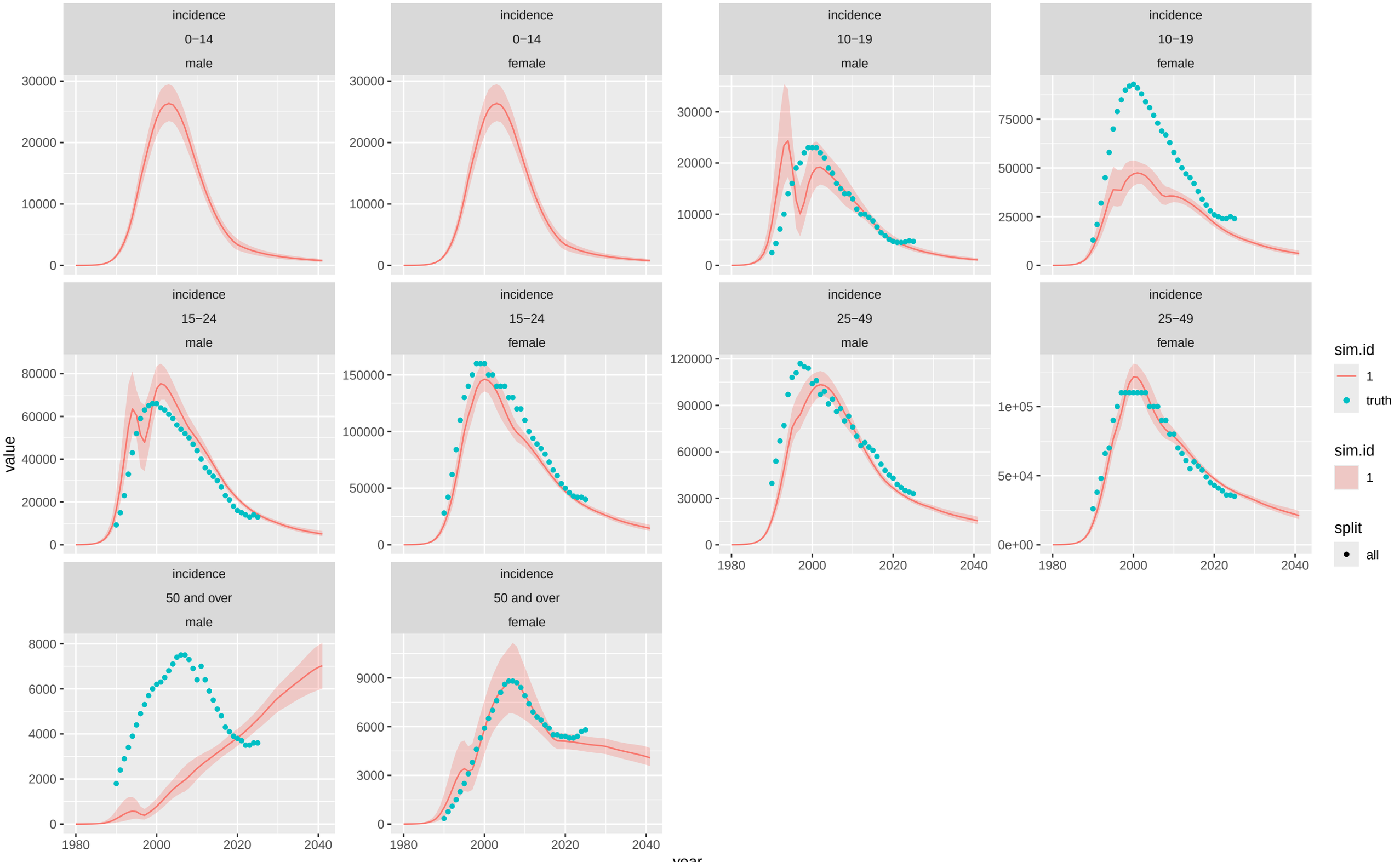
South Africa: prevalence.by.sex (merged Step 2 simset)



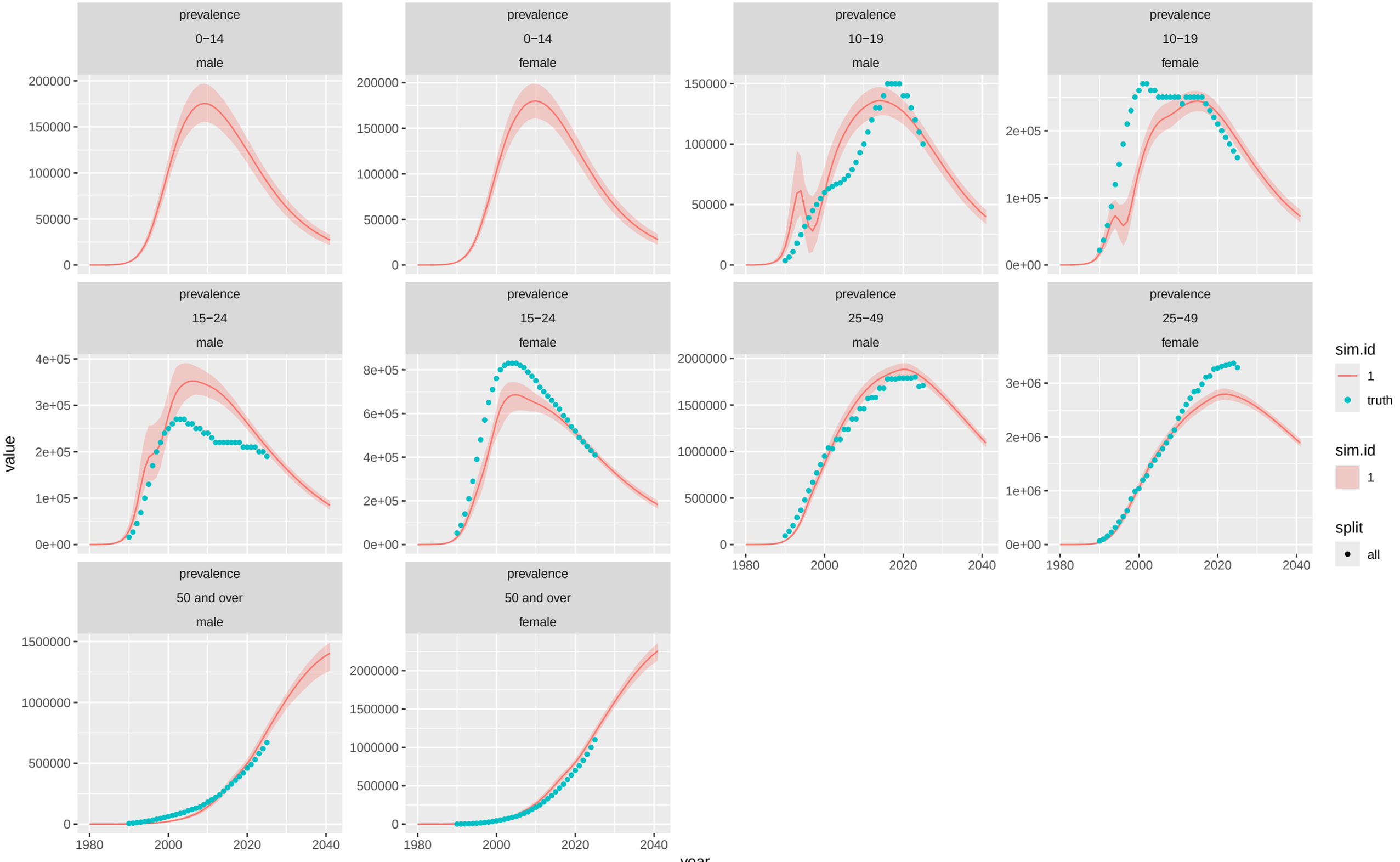
South Africa: hiv.mortality.by.sex (merged Step 2 simset)



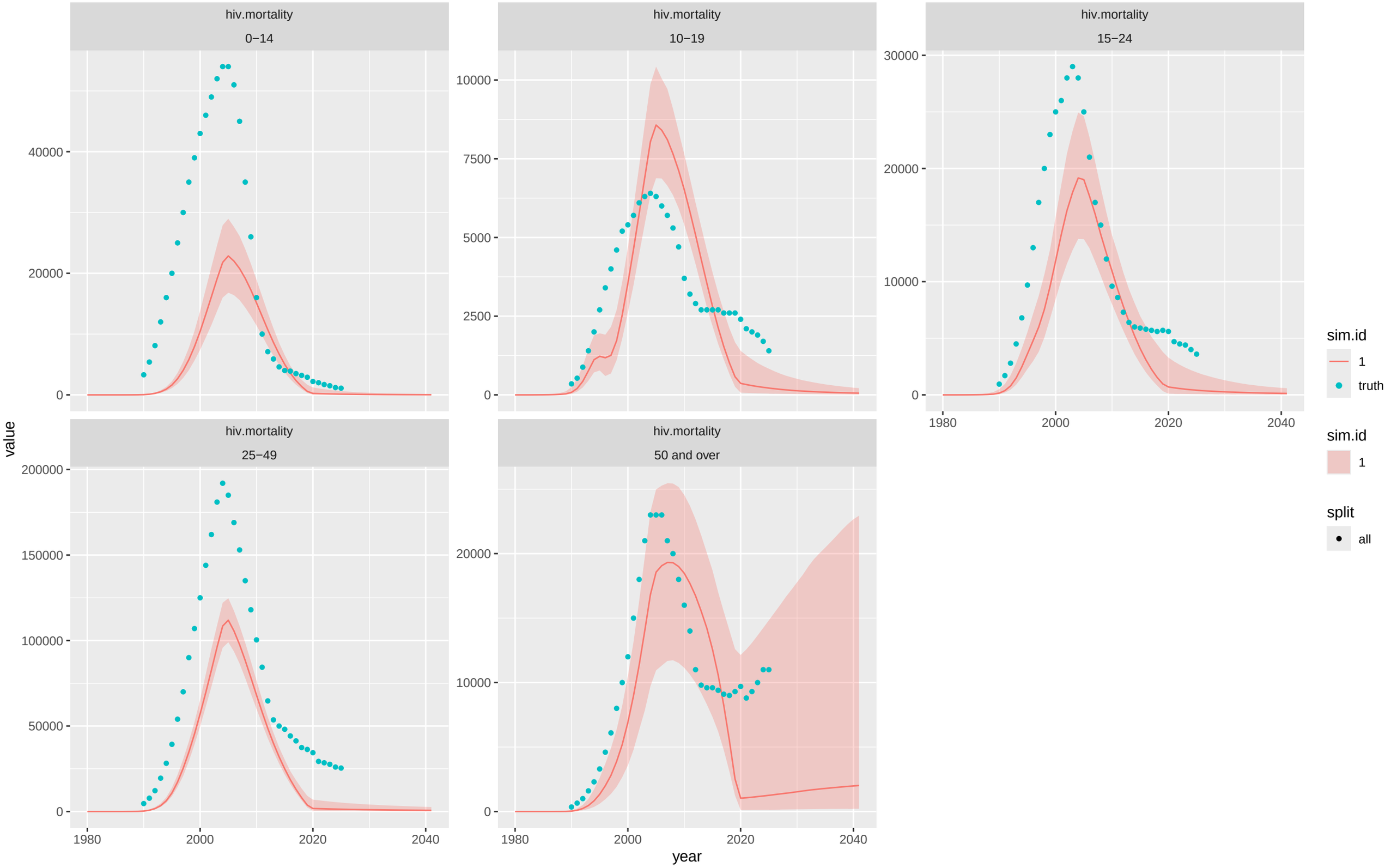
South Africa: incidence.by.age.sex (merged Step 2 simset)



South Africa: prevalence.by.age.sex (merged Step 2 simset)



South Africa: hiv.mortality.by.age (merged Step 2 simset)



South Africa: hiv.mortality.by.age.sex (merged Step 2 simset)